Answer Key - Chemistry - Graduate Level Assessment 20 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** A
   Options: A) 6; B) 9; C) 2; D) 4; E) 10
   Note: The maximum number of phases that can be at equilibrium in a three-component mixture is determined by the phase rule, which states that F = C - P + 2, where F is the degrees of freedom, C is the number of components, and P is the number of phases; thus, setting F to 0 for equilibrium results in a maximum of 6 phases (P = 6) for three components (C = 3).

2. **Answer:** D
   Options: A) 0.175 nm; B) 0.080 nm; C) 0.098 nm; D) 0.123 nm; E) 0.135 nm
   Note: The correct answer of 0.123 nm for the de Broglie wavelength of an electron with a kinetic energy of 100 eV is derived from the de Broglie wavelength formula, = h/p, where the momentum p is calculated using the kinetic energy, leading to the wavelength being approximately equal to 0.123 nm when substituting the appropriate values for Planck's constant and the electron's mass.

3. **Answer:** A
   Options: A) 200 Å; B) 1 Å; C) 0.1 Å; D) 50 Å; E) 500 Å
   Note: The thickness of the oil film can be calculated by dividing the volume of oil (1 cubic millimeter, which is equivalent to 1,000,000 cubic micrometers) by the area it covers (1 square meter or 10,000 square centimeters), resulting in a thickness of 200 angstroms.

4. **Answer:** A
   Options: A) CH 3 COOH; B) CO 2; C) O_2; D) H_2O; E) NH 3
   Note: CH 3 COOH (acetic acid) is a weak electrolyte, not a strong one, because it partially dissociates into ions in water, leading to a lower concentration of ions compared to strong electrolytes that fully dissociate.

5. **Answer:** D
   Options: A) Silicon; B) Boron-doped silicon; C) Pure germanium; D) Arsenic-doped silicon; E) Indium Phosphide
   Note: Arsenic-doped silicon is considered an n-type semiconductor because arsenic, which has five valence electrons, donates extra electrons to the silicon lattice, thus increasing the concentration of electrons available for conduction.

True / False

6. **Answer:** True
   Options: A) True; B) False

Note: The statement is true because the NMR frequency of a nuclide is directly proportional to the strength of the magnetic field, and for 17 O, the calculated frequency at 20.0 T is indeed approximately 115.5 MHz.

7. **Answer:** False
   Options: A) True; B) False
   Note: The statement is false because, according to the first law of thermodynamics, the change in internal energy of the gas is equal to the heat added minus the work done, and in this case, the internal energy change does not correspond to a temperature change of 65 K for 1 mole of a diatomic gas.

8. **Answer:** False
   Options: A) True; B) False
   Note: The expected value of the number of heads when two coins are tossed is 1, as the possible outcomes are 0, 1, or 2 heads, each with probabilities that yield an average of 1 head, not 2.5.

9. **Answer:** True
   Options: A) True; B) False
   Note: The statement is true because an element that displaces hydrogen typically has a valence of -1 or -2, and with an atomic weight of 58.7, it is likely a non-metal that can accept two electrons to form bonds, thus exhibiting a valence of -2.

10. **Answer:** False
    Options: A) True; B) False
    Note: The carbon configuration $1s^2 2s^2 2p^2$ corresponds to 4 distinct quantum states, as each electron in the 2 p subshell can occupy one of the three p orbitals with specific spin orientations, leading to a total of 6 possible arrangements, but only 4 unique states when considering indistinguishable particles.

Long Form Response

11. **Answer:** The kinetic energy (KE) of a 1.0 g particle released from rest near the Earth's surface, under an acceleration of 8.91 m/s^2 after 1.0 s, can be calculated using the formula KE = 0.5 * m * v^2. First, we determine the velocity (v) of the particle after 1.0 s, which is given by v = a * t = 8.91 m/s^2 * 1.0 s = 8.91 m/s. Converting the mass from grams to kilograms, we have m = 0.001 kg. Substituting these values into the kinetic energy formula yields KE = 0.5 * 0.001 kg * (8.91 m/s)^2, resulting in a kinetic energy of approximately 36 mJ. This calculation illustrates the relationship between acceleration due to gravity and the kinetic energy acquired by a particle in free fall when air resistance is neglected.
    Note: correct because it accurately applies the kinetic energy formula and correctly calculates the velocity of the particle after 1.0 seconds using the given acceleration, while also converting the mass from grams to kilograms to ensure consistency in units.

12. **Answer:** The total binding energy of carbon-12 (_6 $C^1$2) is 92.22 MeV, which reflects the energy required to disassemble the nucleus into its individual protons and neutrons. When this total binding energy is divided by the number of nucleons (6), the average binding energy per nucleon is calculated to be approximately 7.68 MeV. This relatively high average binding energy per nucleon suggests that carbon-12 is a stable nucleus, as a greater binding energy generally indicates a stronger nuclear force holding the nucleons together. Consequently, the stability and structure of carbon-12 play a crucial role in its prevalence in the universe, particularly in biological systems, as it serves as a fundamental building block of organic matter.

    Note: correct because it accurately calculates the total binding energy of carbon-12 and its average binding energy per nucleon, demonstrating that the relatively high value of 7.68 MeV indicates a strong nuclear force and contributes to the nucleus's stability.

End of Answer Key